RelCompat3D: Predicate–Geometry Compatibility for Re-Ranking 3D Scene Graph Relations
Supplementary Material

Anonymous submission

Supplementary Method Details

Notation

Table 1 summarizes the notation used in the main paper and the supplement.

Symbol	Meaning
r_i	Relation candidate i
(s_i, o_i)	Ordered subject and object instance identifiers
$p_i,\ a_i$	Predicate label and relation-family label
$k_i^{\mathrm{pair}},\ k_i^{\mathrm{rel}}$	Ordered-pair identity and exact relation-candidate identity
$T_i,\ G_i,\ Z_i$	Predicate semantics, ordered-pair measurements, and source relation score
$q,\ f_q,\ C_i^q$	Compatibility estimator, compatibility logit, and bounded compatibility score
$H_{a_i},\ \mathcal{O}_i$	Applicable transformation group and transformation orbit
$C_i^{\mathrm{tr,q}},\ u_i^q$	Transformation-consistent compatibility and within-family ranking score
$\mathcal{A}_{\mathrm{eval}},\ \mathcal{A}_{\mathrm{rank}}$	Evaluated and re-ranked relation-family sets
$\pi^Z,\ \pi_a^q$	Source ranking and ordered list for family a
$L_K(c),\ Y_c$	Selected top-K candidates and exact ground-truth relations in context c
$D_K,\ N_s, N_u, N_v$	Status-assigned candidates and satisfied, uncertain, and violated counts
$\mathcal{P},\ \ell_i^q$	Linked training pairs and compatibility logit used by the pairwise loss

Table 1: Notation used in the RelCompat3D method and evaluation.

Model and Preprocessing Details

Open3DSG. Open3DSG uses the official non-averaged BLIP checkpoint `epoch=13-step=13104.ckpt`, selected by train/dev validation loss. Public preprocessing produces candidates for 533/548 contexts; the main evaluation retains the full 548-context ground-truth universe and treats the 15 contexts without candidates as empty sets. For sensitivity, the eligible-only denominator contains 3,899 relations, and a documented recovery covers all 548 contexts by lowering the visible-object minimum to two and regenerating views for two scans.

SGFN. We use the released `SGFN_full_l160` model from the SceneGraphFusion 3DSSG implementation, with its 160-object, 26-relation checkpoint on the same evaluation scans and denominator.

Predictor	Candidates	Minimum	Maximum
VL-SAT	110,424	5.30×10^{-22}	0.9954
Open3DSG	79,722	0.6394	0.9281
SGFN	110,424	4.61×10^{-20}	0.5846

Table 2: Observed ranges of source relation scores for candidates in the re-ranked proximity and vertical-order families. No score is negative or exactly zero.

Observed ranges of source relation scores. Table 2 reports the source relation scores observed among the proximity and vertical-order candidates evaluated by the primary re-ranking procedure. All scores are nonnegative, so multiplication by compatibility does not reverse the source-score ordering through a sign change. These are observed evaluation ranges rather than calibrated probability intervals. Open3DSG also has nonnegative scores before family routing, with an all-candidate range of $[0.5772, 0.9707]$.

Compatibility Estimation and Family-Aware Re-Ranking

Guarantees and Target Construction.

Proposition 1 (Exact transformation consistency) *Let a finite transformation group H_a act jointly on the predicate and geometry of an ordered pair, and define $C_a^{\mathrm{tr}}(x) = |H_a|^{-1} \sum_{g\in H_a} C_a(gx)$. Then $C_a^{\mathrm{tr}}(hx) = C_a^{\mathrm{tr}}(x)$ for every $h \in H_a$.*

Proof. For any $h \in H_a$, $C_a^{\mathrm{tr}}(hx) = |H_a|^{-1} \sum_{g\in H_a} C_a(ghx)$. Right multiplication by h is a bijection on the finite group, so re-indexing $g' = gh$ gives $C_a^{\mathrm{tr}}(hx) = C_a^{\mathrm{tr}}(x)$. The two-element proximity and vertical actions yield the identities in the main paper, and the singleton support/contact action is the identity.

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Family	Intervention and acceptance rule	Relation to verifier
Proximity	Choose directed same-context pairs in descending normalized XY distance; require distance $\geq$ 2.5 and zero projected overlap.	Shares distance/overlap primitives and the 2.5 satisfied boundary; the verifier declares violation only at ≥ 3.5.
Vertical	On the same directed pair, replace `higher than` with `lower than` or conversely; require the positive direction to have $\lvert\Delta z\rvert \geq 0.25$ m and normalized $\lvert\Delta z\rvert \geq 0.15$.	Shares both signed primitives and thresholds.
Support/ contact	Replace the subject or object within context; reject known support/contact pairs in either direction and floor subjects; require any two of normalized XY distance ≥ 2.0, zero overlap, and vertical gap $\geq$ 0.30 m.	Shares these OBB primitives and thresholds; point-subtype evidence used by the reported verifier is not used to generate negatives.

Table 3: Counterfactual-negative construction. Normalized XY distance uses the mean ordered-pair XY diagonal. Zero overlap requires both directed projected-overlap ratios to be at most 10^{-9}.

Family-sequence preservation. For each family a, the algorithm constructs the ordered list π_a defined in the main paper. At position j, it selects the next unselected candidate from π_{a_j}; hence the output family sequence is exactly $(a_1, \ldots, a_n)$. The support/contact list follows the source ranking, so the first K positions contain exactly the same ordered support/contact subsequence as the source prefix. Therefore the support/contact selections and their order are identical at every reported K.

Prefix utility optimality. Fix a prefix length K, and let $n_a(K)$ be the number of positions assigned to family a in the source prefix. Every feasible ranking that preserves these family counts must select $n_a(K)$ candidates from each family. Because each re-ranked π_a is sorted by the corresponding estimator's u_i, each RelCompat3D estimator selects the $n_a(K)$ highest-utility candidates within that family. If another feasible selection omits one of these candidates in favor of a lower-utility candidate from the same family, exchanging the two increases the summed within-prefix utility. The support/contact subsequence is fixed, so it does not affect this comparison.

Counterfactual target construction. Ground-truth relations in the evaluated families provide positives. Negatives are generated within the same scene context and linked to their source positive; no source relation score, predictor identity, or evaluation candidate is used. Table 3 gives the complete construction policy.

Construct-dependence boundary. Table 4 makes the information boundary explicit. Training examples come only from the training split, so evaluation candidate rows, source relation scores, and primary verifier-status labels do not enter target construction. The counterfactual rules and primary verifier nevertheless share some OBB-derived measurements and family thresholds. The point- and mesh-based audit removes that OBB overlap but retains the reconstructed evaluation scenes and relation ontology.

At most two negatives are emitted per positive, 200 per context–family, and three times the number of positives per family. These caps limit repeated easy negatives; they do not equalize family counts. The export contains 28,977 positives and 37,477 negatives. Total rows by family are 23,008 proximity, 5,488 vertical, and 37,958 support/contact. The explicit overlap above is why Violation computed by the rule-based geometry verifier is reported as a verifier-derived measure rather than independent physical-validity ground truth; the feature-removal analysis tests, but does not eliminate, this construct dependence.

Estimator and Optimization Details. RelCompat3D-Linear fits one logistic model per evaluated relation family. Numeric features use means and standard deviations estimated from the training split; missing values use the corresponding training mean. Inputs are predicate indicators T, predicate-independent geometric measurements of the ordered pair G, and the two predicate-signed height interactions defined in the main paper. The family label selects the head and is not repeated as an input. The source relation score, source rank, predictor identity, and object class are excluded. Transformation-consistent augmentation preserves the proximity predicate under endpoint swap; vertical order swaps endpoints and applies the inverse predicate. Linked counterfactuals receive the fixed margin loss in the main paper. Support/contact receives the pairwise loss but no blanket endpoint transform.

RelCompat3D-MLP fits one shared multilayer perceptron (MLP) with a single two-unit ReLU hidden layer and a predicate-linear skip connection. Its 29 inputs comprise family and predicate indicators, the same 17 geometric measurements, the two signed-height interactions, and the sum of the directional overlap ratios. The family indicator is nonconstant because this estimator shares parameters across families. It has 69 parameters and excludes the source relation score and predictor identity.

The 60,208 training rows yield 33,961 linked pairs, and transformation-consistent augmentation produces 86,032 optimization rows for both estimators. Linear uses 800 deterministic full-batch steps at learning rate 0.2; MLP uses 120 steps at learning rate 0.02. Both use the same pairwise weight, margin, ℓ_2 coefficient, and fixed seed policy. The three linear family models contain 21–23 parameters. Each reported condition uses one fit with a frozen seed; the paired intervals quantify scan resampling rather than retraining variation.

Supplementary Experiments

Experimental Setup

Implementation and Development Diagnostics.

Information	Training construction	Primary verifier	Point/mesh audit
Evaluation candidate rows	No	Yes	Yes
Source relation score	No	No	No
Primary verifier-status labels	No	Output	No
OBB-derived measurements	Some	Yes	No
Point/mesh measurements	No	No	Yes
Evaluation scene identities	No	Yes	Yes
Relation ontology	Yes	Yes	Yes

Table 4: Information used by target construction and the two evaluation routes. "Some" denotes the OBB primitives and family thresholds shared by counterfactual construction and the primary verifier. The point/mesh audit is an alternative geometric construct, not independent physical-validity ground truth.

Predictor	Contexts	Scored candidates	Total (s)	ms/context
VL-SAT	548	110,424	2.445	4.461
Open3DSG	533	79,722	1.808	3.391
SGFN	548	110,424	2.453	4.476

Table 5: CPU re-ranking cost for RelCompat3D-Linear. Open3DSG has 533 nonempty public-prediction contexts; its 15 empty official contexts require no scoring or sorting.

Family	Train	Dev	Params	Pairs	AUROC	Brier
Proximity	20,902	2,106	21	9,562	1.0000	.0009
Vertical order	4,922	566	22	1,644	.9982	.0079
Support/contact	34,384	3,574	23	22,755	.9933	.0256

Table 6: RelCompat3D-Linear training and development discrimination. Labels comprise ground-truth-positive/counterfactual pairs, not human physical-validity judgments.

Computational cost. We benchmark the fixed RelCompat3D-Linear CPU implementation with one process and one thread on an Intel Core Ultra 7 265KF with 62 GiB RAM, Linux 6.17, and Docker 29.6.1. The Debian Bookworm container pins Python 3.11.9, NumPy 1.26.4, and Pillow 10.4.0; source inference used an NVIDIA GeForce RTX 5090 with 32 GiB memory. Timing starts from preloaded candidates and includes compatibility, transformation averaging, sorting, and assembly, but excludes source inference, reconstruction, geometry association, file parsing, metrics, and bootstrap. Table 5 reports medians over five runs after one warm-up. Exact seeds, commands, and model/protocol hashes are in the anonymous code artifact.

The three stored linear heads contain 66 parameters; the primary proximity/vertical path evaluates 43 of them and adds no fitted fusion parameter. Peak process RSS, including the preloaded candidates, is 366.5 MiB. All five runs preserve the exact family sequence and support/contact order.

Across all 6,246 development rows, AUROC/AUPRC/Brier are .9973/.9973/.0157. On 3,516 linked development pairs, the positive wins 99.23% of the time with mean logit margin 14.14. Correct-predicate compatibility wins on 97.40% of 384 vertical positives. Exact transformation averaging gives a maximum compatibility difference of zero under proximity endpoint swap and under joint endpoint swap and inverse-predicate transformation for vertical order, on development and evaluation rows from all three predictors. The shared nonlinear estimator obtains development AUROC/AUPRC/Brier of .9904/.9926/.0225. Its transformation-consistent compatibility also changes monotonically in all 512 proximity and 512 vertical intervention cases used by the point- and mesh-based audit.

Row-Level Regeneration Check. We export a deterministic derived-row bundle for the 548-context evaluation universe. It contains pseudonymized candidate and context identities, predictor, predicate and family labels, source score, Linear and MLP compatibility, paper ranking positions, exact-match membership, and primary and point/mesh verifier statuses. Original scan and instance identifiers, object categories, and raw geometric measurements are excluded. From these rows, one Docker command regenerates Tables 1–3, the data and a verification rendering for Figure 3, and a cell-by-cell comparison with the frozen paper values. The completed run checks 291 cells with maximum absolute error zero at tolerance 10^{-12}. The anonymous artifact includes the exporter, reproducer, schema, compact regenerated outputs, and hash manifests. Redistribution of the source-derived row bundle itself remains conditional on the dataset terms, so it can instead be regenerated from licensed external inputs.

Results

Component and Feature-Removal Analyses.

Matched component diagnostics. Tables 7–9 isolate the linked pairwise loss and transformation averaging for both compatibility estimators. Within each estimator, all conditions retain the same rows, features, optimizer, source scores, and family-aware ranking procedure. No pairwise removes only the linked pairwise term while retaining relation-preserving augmentation and transformation averaging. No averaging uses the fitted full estimator but scores compatibility without inference-time transformation averaging.

The pairwise term raises the held-out positive-win rate by 0.085 percentage points for Linear and 0.057 points for MLP. It also increases the mean Linear margin and reduces its softplus margin loss from .03607 to .03534. The MLP margin summaries are mixed: its mean margin and softplus loss do not improve. The term is therefore treated as a training

Predictor	Condition	$K = 50$	$K = 100$
VL-SAT	Linear full	.9277/.0197	.9658/.0295
	Linear no pairwise	.9277/.0198	.9658/.0297
	Linear no averaging	.9277/.0197	.9660/.0295
	MLP full	.9272/.0189	.9650/.0296
	MLP no pairwise	.9275/.0180	.9648/.0280
	MLP no averaging	.9275/.0189	.9645/.0296
Open3DSG	Linear full	.4418/.0342	.5685/.0324
	Linear no pairwise	.4418/.0342	.5692/.0324
	Linear no averaging	.4413/.0342	.5692/.0324
	MLP full	.4670/.0413	.5989/.0371
	MLP no pairwise	.4600/.0399	.5939/.0362
	MLP no averaging	.4507/.0398	.5735/.0368
SGFN	Linear full	.7450/.0263	.9303/.0350
	Linear no pairwise	.7450/.0264	.9303/.0353
	Linear no averaging	.7450/.0263	.9303/.0350
	MLP full	.7457/.0258	.9288/.0350
	MLP no pairwise	.7452/.0255	.9295/.0330
	MLP no averaging	.7457/.0258	.9300/.0352

Table 7: Matched component removals. Cells are exact-match Recall / verifier-derived Violation reported as proportions. Full, no-pairwise, and no-averaging conditions are matched separately within each estimator.

regularizer with limited, estimator-dependent diagnostic effect, not as the main cause of the aggregate Recall–Violation changes.

Transformation averaging gives zero compatibility error and identical top-K membership under transformed representations for both full estimators and their no-pairwise refits. Without averaging, the minimum top-K Jaccard is .9635 for Linear and .6975 for MLP. Thus averaging supplies the claimed exact consistency guarantee even though its aggregate effect in Table 7 is small.

Training-seed robustness. Table 10 separates fitting variation from the scan-resampling intervals in the main paper. Five model-seed identifiers were fixed before evaluation, while the constructed rows and linked-pair identities were held fixed. The active MLP seed, 20260714, was included but not selected from this analysis. Linear uses deterministic zero initialization and full-batch optimization, so all five executions produce the same model hash.

Linear is exactly invariant across the five executions. For MLP, every predictor–K cell retains both favorable Source directions in all five runs except VL-SAT at $K = 50$: seed 20260718 lowers Recall by one exact relation, from 92.724% to 92.699%, while lowering Violation from 2.675% to 1.821%. The active fit is unchanged. These results support low fitting variation but do not justify a claim of seed-uniform Pareto improvement.

The factor ablation uses the same split and estimator class in a separate pooled analysis; because one head spans all families there, its semantic input includes predicate and family identifiers. At development, semantic-only, predicate-independent G-only, additive semantics+G, and interaction $T \times G$ obtain AUROC .6124/.8809/.9193/.9822 and Brier .2370/.1363/.1061/.0495. Shuffled and wrong-pair geometry reduce the intended reliability signal. These results verify factor use and the implemented transformation identities; they do not convert the constructed target into independent physical-validity ground truth.

Every feature-removal model is refitted on the 1,061 training scans with the same linked positive–counterfactual objective, exclusion of the source relation score, and exact transformation averaging. The exact-scalar condition retains development AUROC of .99999 for proximity and .99794 for vertical order. Removing all related distance or vertical measurements lowers these to .9512 and .7089; alternative evidence yields .8763 and .5531. At $K = 50$, the strict conditions retain a clear Open3DSG improvement but not the full Violation reduction on VL-SAT or SGFN. At $K = 100$, all feature-removal conditions improve both reported metrics across all three predictors. The results show that performance does not depend solely on a single verifier input, but they do not eliminate broader construct dependence on correlated geometry. The main paper reports the matched controls under the active family-aware ranking procedure; the separate all-family product and pooled comparison appears in Table 22.

Point- and Mesh-Based Consistency Audit. The primary audit covers the proximity and vertical-order families reranked by RelCompat3D. It does not assess support/contact, whose candidates remain in source order and require a separate contact construct. Point measurements use instance vertices: a symmetric 10th-percentile nearest-surface distance normalized by a robust point radius for proximity, and the difference between median vertical coordinates for vertical order. Mesh measurements use area-weighted triangle centroids from the reconstructed instance surfaces and an area-derived scale. They do not read OBB measurements, predictor identity, source relation scores, compatibility scores, or primary verifier labels.

All thresholds are fixed using positive relations from the 1,061-scan training split. Point measurements require 49 samples per endpoint and use normalized proximity boundaries 0.6217/1.1645 and a 0.0941 m absolute vertical margin; mesh measurements require 56 samples and use 0.7230/1.4521 and 0.0826 m. The lower and upper proximity boundaries are the training-positive 90th and 99th percentiles, and the vertical margin is the 10th percentile of the absolute signed separation. An agreement-based label is satisfied or violated only when the point- and mesh-based labels agree; measured disagreements are uncertain. Candidates without both measurements do not enter the audit Violation denominator.

The audit has point and mesh data for all 157 evaluation scans. At $K = 50/100$, agreement-based measured/decidable coverage for Linear is 95.5/71.0 and 95.2/73.8% on VL-SAT, 98.2/83.7 and 98.1/82.5% on Open3DSG, and 95.8/79.9 and 95.8/77.6% on SGFN. The corresponding MLP values are 95.6/71.3 and 95.4/74.1%, 98.3/85.7 and 98.1/82.9%, and 95.8/80.2 and 95.9/78.1%. Measured disagreements remain uncertain and enter the denominator but not the violation numerator.

For each estimator, agreement-based Violation decreases

Estimator	Condition	Pairs	Positive win	Mean margin	P05	Median	P95
Linear	Full	3,516	.9926	14.1413	4.1397	13.3462	27.8989
Linear	No pairwise	3,516	.9918	13.6145	4.0170	12.7360	26.9527
MLP	Full	3,516	.9932	12.1491	3.9317	11.6214	21.8631
MLP	No pairwise	3,516	.9926	12.6489	3.7400	12.0326	22.5406

Table 8: Held-out linked-pair ordering diagnostics. Margins are positive-minus- counterfactual compatibility logits on the 117-scan development split after transformation averaging. P05 and P95 are margin percentiles.

<table>
<tr><th>Estimator</th><th>Condition</th><th>Family</th><th>Mean</th><th>P95</th><th>Max</th><th>Min Jaccard</th><th>Min exact-context</th></tr>
<tr><td>Linear</td><td>Full</td><td>Both</td><td>.0000</td><td>.0000</td><td>.0000</td><td>1.0000</td><td>1.0000</td></tr>
<tr><td>Linear</td><td>No pairwise</td><td>Both</td><td>.0000</td><td>.0000</td><td>.0000</td><td>1.0000</td><td>1.0000</td></tr>
<tr><td>Linear</td><td>No averaging</td><td>Proximity</td><td>.0150</td><td>.0817</td><td>.4532</td><td rowspan="2">.9635</td><td rowspan="2">.8266</td></tr>
<tr><td>Linear</td><td>No averaging</td><td>Vertical</td><td>.0014</td><td>.0080</td><td>.0192</td></tr>
<tr><td>MLP</td><td>Full</td><td>Both</td><td>.0000</td><td>.0000</td><td>.0000</td><td>1.0000</td><td>1.0000</td></tr>
<tr><td>MLP</td><td>No pairwise</td><td>Both</td><td>.0000</td><td>.0000</td><td>.0000</td><td>1.0000</td><td>1.0000</td></tr>
<tr><td>MLP</td><td>No averaging</td><td>Proximity</td><td>.1684</td><td>.5805</td><td>.9370</td><td rowspan="2">.6975</td><td rowspan="2">.4617</td></tr>
<tr><td>MLP</td><td>No averaging</td><td>Vertical</td><td>.0402</td><td>.2896</td><td>.9157</td></tr>
</table>

Table 9: Transformation diagnostics. Mean, P95, and Max are the worst source-specific summaries of the absolute compatibility difference under the applicable endpoint/predicate transformation. The last columns give the minimum transformed-view top-K membership consistency over all predictors and reported K.

in 14 of the 15 predictor–K settings and is tied for SGFN at $K = 5$; it never increases. The 14 non-tied paired intervals are strictly below zero. This audit therefore supports the direction of the primary result under alternative geometric measurements, while Table 4 states the remaining shared-scene and shared-ontology boundary.

As a mechanism test, 512 synthetic interventions per family translate one endpoint away from the other for proximity or across the vertical decision boundary. Both compatibility estimators respond monotonically in all proximity and vertical cases. The point and mesh measurements respond monotonically in 96.29/94.73% of proximity cases and 100% of vertical cases; distance and height quantiles can be non-monotone under translation for non-convex objects. The audit therefore reduces dependence on the exact OBB rules used in the primary metric, but both point- and mesh-based estimates still use the same reconstructed 3RScan geometry and relation ontology. They are not independent physical-validity ground truth.

Qualitative Pair Analysis. The Open3DSG context used for the main-paper teaser also contains a complementary promotion case. The exact-label relation "desk close by chair" is verifier-satisfied, with an XY center distance of $0.436\,\mathrm{m}$. RelCompat3D-Linear moves this candidate from source rank 81 to rank 30, bringing it into the top-50. This case records a promotion outcome without implying a direct exchange with the vertical-order candidate, because the two candidates belong to different relation families.

Figure 1 links each ordered-pair projection to the measured evidence and ranking outcome. The proximity and vertical examples are demoted by both compatibility estimators. The support/contact example is retained in source order because that family is outside the re-ranking scope.

Counterfactual-Policy Sensitivity. We regenerate the training and internal-development targets and refit the proximity and vertical heads under nine one-factor-at-a-time conditions. Each variant recomputes normalization from the 1,061 training scans; no final- validation row enters target generation, normalization, or fitting. The default refit reproduces the main model and metrics exactly. Support/contact is kept in source order in every condition.

Across all settings, the maximum absolute change from the default is .0020 Recall and .0011 Violation at $K = 50$, and .0035 Recall and .0020 Violation at $K = 100$. Every setting retains lower Violation and no lower Recall than Source for all three predictors at $K = 50$ and $K = 100$. This reduces dependence on a particular threshold or negative count, but does not make the constructed target independent of the verifier.

Estimator, Fusion, and Routing Sensitivities.

Source-score mapping sensitivity. The source ranking is invariant to a strictly monotonic score mapping, whereas the product $Z_i C_i^{\mathrm{tr,q}}$ need not be. We therefore apply a frozen grid of mappings before product scoring without selecting a replacement mapping from evaluation results. The smooth grid contains

$$q_\gamma(Z) = Z^\gamma, \quad \gamma \in \{0.5, 2, 4\},$$

and

$$\begin{aligned} q_T(Z) &= \sigma\bigl(\mathrm{logit}(\bar{Z})/T\bigr)\,, \\ \bar{Z} &= \mathrm{clip}(Z, \epsilon, 1-\epsilon), \\ \epsilon &= 10^{-6}, \qquad T \in \{0.5, 2\}. \end{aligned}$$

The temperature mappings are monotonic scale stresses and do not treat the Open3DSG cosine score as a calibrated probability. A separate context-and-family percentile mapping as-

Predictor	Estimator	Statistic	$K = 5$	$K = 10$	$K = 20$	$K = 50$	$K = 100$
VL-SAT	Linear	Mean	42.07/.15	63.39/.57	80.82/1.14	92.77/1.97	96.58/2.95
	Linear	SD	.00/.00	.00/.00	.00/.00	.00/.00	.00/.00
	MLP	Mean	42.08/.15	63.43/.52	80.89/1.10	92.72/1.87	96.52/2.92
	MLP	SD	.04/.00	.05/.01	.02/.05	.02/.11	.01/.27
Open3DSG	Linear	Mean	3.73/.94	11.38/2.33	23.62/3.13	44.18/3.42	56.85/3.24
	Linear	SD	.00/.00	.00/.00	.00/.00	.00/.00	.00/.00
	MLP	Mean	3.71/2.27	11.71/3.28	24.18/3.65	45.86/3.72	59.30/3.50
	MLP	SD	.02/1.38	.07/.87	.32/.47	.42/.23	.45/.13
SGFN	Linear	Mean	31.17/2.37	39.75/3.47	49.14/2.97	74.50/2.63	93.03/3.50
	Linear	SD	.00/.00	.00/.00	.00/.00	.00/.00	.00/.00
	MLP	Mean	31.17/2.37	39.75/3.47	49.19/2.95	74.60/2.57	92.96/3.44
	MLP	SD	.00/.00	.00/.00	.02/.00	.03/.05	.06/.30

Table 10: Five-seed fitting robustness. Cells are Recall / verifier-derived Violation in percent. Mean and population standard deviation (SD) are computed over five predeclared fitting executions; they are separate from scan resampling uncertainty.

signs descending scores from one to zero within each context and family.

Table 16 summarizes all five reported K values. A favorable cell has Recall no lower and verifier-derived Violation no higher than Source. The smooth grid yields 75/75 favorable Linear cells and 74/75 favorable MLP cells. The only smooth-grid exception is VL-SAT MLP at $K = 50$ under Z^4, where Recall changes by -0.025 percentage points, its paired interval includes zero, and Violation decreases. The percentile stress produces two Linear and four MLP Recall losses, none larger than 0.227 percentage points, without increasing Violation. Across the complete grid, including percentile mappings, the minimum pair-weighted Kendall correlation with identity-product ranking is 0.810 and the minimum $K = 50$ micro-Jaccard overlap is 0.815. These results show bounded sensitivity over the tested mappings, not score-scale invariance.

Closest simple baseline. Positive-density is a non-learned continuous compatibility baseline fitted only from training-split positive geometry. It uses predicate-specific medians and interquartile ranges, transformation averaging, the same product score, and the same family-aware re-ranking rule. It does not use evaluation-verifier labels or constructed counterfactual negatives. Table 17 reports all K values. At $K = 50$, both learned estimators Pareto-dominate Positive-density for all three predictors. Over all 15 predictor–K settings, Linear dominates it in 12 settings and trades Recall against Violation in three. MLP dominates it in 12, has a trade-off in two, and is dominated only for Open3DSG at $K = 5$.

Direct-verifier diagnostics. Hard-tail places verifier-violated candidates after non-violated candidates inside each re-ranked family. Hard-drop removes verifier-violated candidates and keeps the source order of the remaining rows. Both read evaluation verifier labels and are therefore non-deployable diagnostics rather than comparable baselines. Table 18 reports their all-K behavior. Hard-drop obtains zero primary Violation by construction, but it returns fewer than the available top-K rows for Open3DSG at $K = 50, 100$ and for VL-SAT/SGFN at $K = 100$. Its retained uncertain rows also prevent zero primary Violation from implying complete geometric evidence.

Compatibility estimators and matched fusion. RelCompat3D-Linear uses family-specific linear heads, whereas RelCompat3D-MLP uses one shared MLP with a single two-unit ReLU hidden layer and a predicate-linear skip connection. They share the constructed training target, linked-pair objective, transformation averaging, exclusion of source relation scores and predictor identities, the multiplicative within-family ranking score, and family-aware re-ranking. The MLP input includes a family indicator because its parameters are shared, plus an overlap-sum interaction derived from the same directional overlap measurements. Both are fitted once and applied to all three predictors without target-specific refitting. Rank-average fusion (RankAvg) and reciprocal rank fusion (RRF) use the linear compatibility scores and the same family-aware ranking procedure.

Let q_i^Z and q_i^C denote descending within-context percentile ranks of the source relation score and compatibility, and let r_i^Z, r_i^C be their one-indexed ranks. Rank-average and fixed-constant Reciprocal Rank Fusion use

$$S_i^{\mathrm{rank}} = \tfrac{1}{2}(q_i^Z + q_i^C), \qquad S_i^{\mathrm{RRF}} = \frac{1}{60 + r_i^Z} + \frac{1}{60 + r_i^C}. \tag{1}$$

At $K = 50$, RelCompat3D-MLP has higher Open3DSG Recall than RelCompat3D-Linear but also higher Violation. They are nearly tied on VL-SAT and SGFN. Rank fusion can lower Violation at larger K values but incurs larger predictor-dependent Recall losses. No estimator or fusion dominates the joint trade-off. A separate exact-match MLP uses stronger SGFN-specific correctness supervision and is therefore retained only as a source-specific upper comparator, not as an apples-to-apples baseline.

Matched structural controls. Across both reported K values and both compatibility estimators, corrupted predicates and pair geometry increase verifier-derived Violation, while compatibility-only ordering loses substantial Recall on

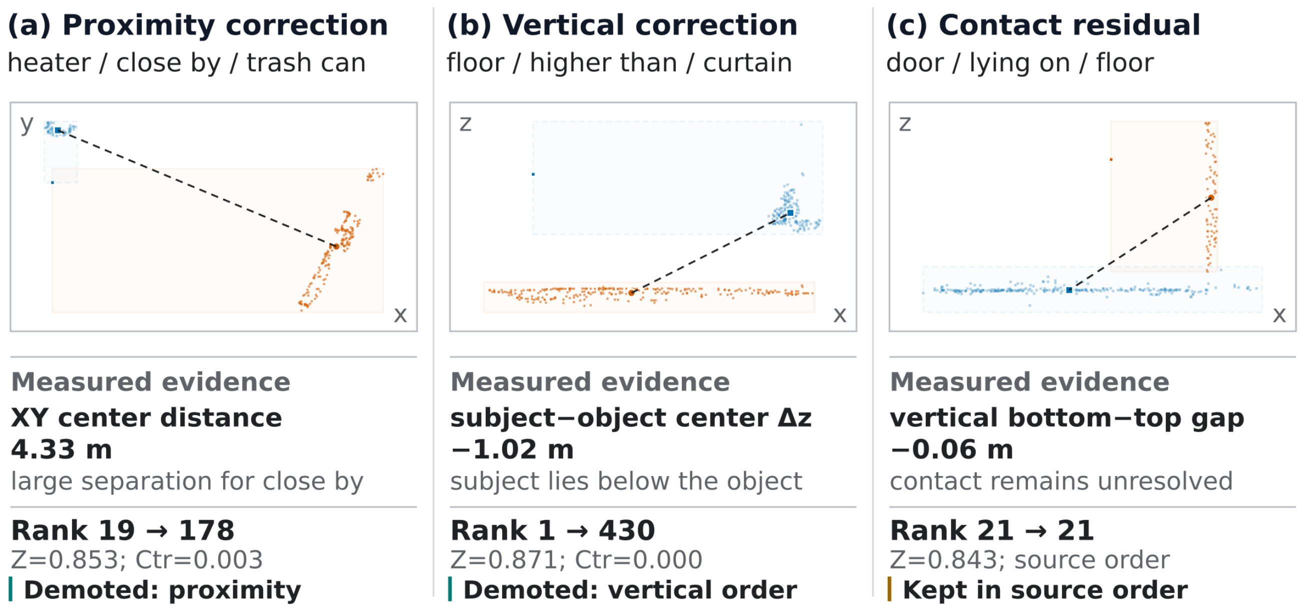

Figure 1: Pair–evidence–outcome analysis. Orange circles and solid boxes identify the subject. Blue squares and dashed boxes identify the object. The first two violations are demoted. The support/contact example requiring contact evidence is kept in source order.

VL-SAT and SGFN. Open3DSG remains more sensitive to source-score scale.

Pooled compatibility ablation. The pooled product fits one compatibility model across all families and does not use the primary family-aware ranking procedure. Compared with Product (all families), it can increase Recall but generally raises Violation, particularly for Open3DSG and SGFN:

Routing-constraint controls. The active *family-slots* route preserves the source family label at every rank position and re-ranks proximity and vertical-order candidates in separate queues. The direct matched control *P/V global* uses the same candidates, compatibility estimates, product scores, support/contact positions, and support/contact identities, but merges proximity and vertical-order candidates into one queue. Table 23 reports this direct test at all five K values.

No route dominates across estimators and K. For Open3DSG, P/V global raises Linear Recall at all five K values. For the MLP, it raises Recall at $K = 5, 10$ but lowers Recall by 2.92 and 3.80 percentage points at $K = 50, 100$, with both paired intervals below zero. At Open3DSG $K = 50$, the MLP control changes the selected proximity/vertical-order counts from 6,295/6,508 to 3,423/9,380 while leaving all 13,833 support/contact selections unchanged. We therefore interpret family slots as a composition-preserving constraint, not an aggregate-optimal route.

Table 24 separates the matched control above from two broader scope comparisons. *Support-order only* preserves only the relative source order of support/contact candidates and permits global cross-family competition. *All families* additionally applies learned compatibility to support/contact and is not a matched test of the family-slot constraint.

Open3DSG coverage sensitivity. The public preprocessing drops a context when fewer than four annotated objects retain view metadata. This accounts for all 15 missing contexts. Table 25 distinguishes the eligible-context evaluation, the conservative full-target main evaluation, and a documented full-coverage recovery.

The conclusion is stable to all three denominator and coverage choices. The main paper uses the conservative full-target public evaluation; recovered results are sensitivity evidence rather than the headline benchmark.

Candidate-Pool Oracle Recall. RelCompat3D can reorder only candidates supplied by each predictor. We therefore measure the fraction of exact ground-truth relations present anywhere in each fixed candidate pool and three diagnostic Recall upper bounds. The *active-route oracle* preserves source family counts and the selected support/contact subsequence while filling proximity and vertical-order slots optimally. The *family-slot oracle* preserves only the source family counts, and the *unconstrained oracle* selects any available exact-match candidates within each context. These are upper bounds, not attainable model results.

Pool coverage is 99.72% for VL-SAT and SGFN but 79.68% for Open3DSG. At $K = 50$, the active-route oracle reaches 96.73%, 63.72%, and 86.05% Recall, respectively. The gap from the observed rankings shows remaining within-route scoring headroom, while the lower Open3DSG pool coverage also quantifies errors that fixed-candidate reranking cannot recover.

Paired Scan-Level Intervals. The analysis resamples the 157 scan identifiers 1,000 times and carries every context of

Predictor	Condition	$K = 50$	$K = 100$
VL-SAT	Source	.9272/.0268	.9635/.0476
	Full model	.9277/.0197	.9658/.0295
	Exact verifier scalar removed	.9280/.0199	.9660/.0300
	Related measurements removed	.9275/.0268	.9653/.0468
	Alternative evidence only	.9280/.0268	.9642/.0471
Open3DSG	Source	.4043/.1387	.5111/.1242
	Full model	.4418/.0342	.5685/.0324
	Exact verifier scalar removed	.4381/.0342	.5687/.0325
	Related measurements removed	.4119/.0954	.5264/.0955
	Alternative evidence only	.4104/.1194	.5327/.1188
SGFN	Source	.7402/.0385	.9235/.0630
	Full model	.7450/.0263	.9303/.0350
	Exact verifier scalar removed	.7465/.0268	.9300/.0358
	Related measurements removed	.7467/.0388	.9298/.0616
	Alternative evidence only	.7462/.0386	.9277/.0625

Table 11: Feature-removal analysis. Cells are exact-match Recall / verifier-derived Violation under the same family-aware ranking procedure. Exact verifier scalar removed omits the normalized scalar consumed by the verifier. Related measurements removed omits all directly related distance or vertical measurements. Alternative evidence only retains projected overlap for proximity and horizontal-distance/overlap context for vertical order.

a sampled scan together. The ranking, model, and point estimates remain unchanged; shared bootstrap indices are used across conditions. A paired 548-context bootstrap gives the same directions and is retained as a finer-grained sensitivity.

For both compatibility estimators, all Recall point changes are non-negative. Open3DSG Recall intervals are strictly positive at every K, and the SGFN $K = 50/100$ intervals are positive. VL-SAT support depends on the estimator and K. Every Violation interval is strictly below zero except the tied SGFN $K = 5$ point and its $K = 10$ interval, which reaches zero. Thus the point-estimate trend is broad, while statistical support remains predictor- and K-dependent.

External Transfer Stress Test. We apply the 3DSSG-trained compatibility model and family-aware ranking rule to the official ReplicaSSG test scenes using FROSS predictions. No target-specific parameter fitting, normalization, or imputation is performed in this evaluation. The exact mapping covers `near`, `above`, and `under`; support/contact is excluded because the ontologies do not provide an exact mapping. We report this as a transfer stress test rather than evidence of dataset-level generalization.

Paired scene-bootstrap intervals support joint improvement at $K = 10$ (ΔR [.0048,.1467], ΔV [$-$.1000,$-$.0091]) and $K = 50$ (ΔR [.0179,.0811], ΔV [$-$.0683,$-$.0185]). The $K = 20$ Recall interval touches zero, while $K = 5$ is inconclusive. At $K = 100$, heavy quantization of the source relation score makes multiplicative re-ranking nearly inert, and candidate support caps exact-match Recall at 44.19%. The stress test therefore supports K-dependent transfer, not dataset-level generalization.

Family Composition. Table 30 reports each family's slice inside the actual global top-100 ranking. Deltas compare RelCompat3D-Linear with Source using shared paired context-bootstrap indices.

Support/contact is exactly unchanged because the ranking procedure preserves both its selection and the source family sequence at every K. Vertical order drives most of the aggregate reduction. The result is consequently a scoped operating-point improvement, not a uniform family claim. The same direction holds across the reported K values. Full machine-readable family slices and simultaneous intervals accompany the code artifact.

Verifier-Uncertainty Sensitivity. The reported $\mathrm{V@K}_{all}$ includes uncertain candidates in its denominator but does not label them satisfied. We therefore recompute each ranking with $\mathrm{V@K}_{dec}$ over satisfied/violated candidates, uncertainty rate $U@K$, and an uncertain-as-violation variant $\mathrm{V@K}_{u \to v}$ that counts every uncertain candidate as a violation. Table 31 gives the $K = 100$ comparison.

RelCompat3D-Linear-minus-Source changes in V_{dec} are $-.0277$ for VL-SAT, $-.1578$ for Open3DSG, and $-.0440$ for SGFN; their paired 95% intervals all exclude zero. Uncertainty-rate changes are +.0009, $-.0076$, and +.0071, whereas uncertain-as-violation changes are $-.0173$, $-.0993$, and $-.0209$. Thus both decidable-only and uncertain-as-violation V are lower for all three predictors even though uncertainty does not decrease uniformly. Across both estimators, primary, decidable-only, and uncertain-as-violation V are non-increasing relative to Source in all 30 predictor–K settings. The Hard-drop diagnostic has zero Violation when the denominator includes all verifier statuses, but it retains uncertain candidates and therefore has uncertain-as-violation V of .3712/.4722/.4042. Thus zero measured Violation does not imply reliable or complete selection.

Predictor	K	Point			Mesh			Agreement		
		Source	Linear	MLP	Source	Linear	MLP	Source	Linear	MLP
VL-SAT	5	1.36	0.93	0.85	2.19	1.43	1.18	0.93	0.51	0.42
	10	5.70	5.11	4.67	5.30	4.65	4.31	3.61	3.06	2.79
	20	10.45	9.47	9.25	9.46	8.55	8.27	7.27	6.36	6.17
	50	20.11	18.11	17.74	19.50	17.51	17.19	16.43	14.34	14.05
	100	29.28	24.89	24.90	29.44	25.09	25.12	26.00	21.50	21.57
Open3DSG	5	83.18	1.48	17.16	83.89	1.62	17.52	83.20	1.26	16.84
	10	65.96	2.18	13.64	65.94	2.10	13.83	65.18	1.60	13.18
	20	55.29	3.89	9.80	55.07	3.93	10.07	53.80	2.90	9.17
	50	47.76	6.28	10.02	47.85	6.57	10.27	45.96	4.90	8.84
	100	42.65	10.25	13.99	43.05	10.77	14.38	40.87	8.65	12.32
SGFN	5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	10	3.06	2.18	1.53	3.02	1.94	1.72	2.42	1.54	1.32
	20	6.68	5.17	4.61	6.90	5.44	4.92	5.33	3.89	3.55
	50	14.05	10.73	10.09	13.61	10.26	9.71	11.56	8.13	7.63
	100	24.45	19.05	18.89	24.33	18.96	18.75	21.55	15.98	15.88

Table 12: Violation percentages derived from point- and mesh-based measurements across all reported K values. Point and mesh use distinct geometric summaries. Agreement requires their binary labels to agree. This audit is a separate construct and is not directly comparable to primary Violation.

Predictor	K	Point ΔV (95% CI)	Mesh ΔV (95% CI)	Agreement ΔV (95% CI)
VL-SAT	5	-0.42 [-0.83, -0.09]	-0.76 [-1.37, -0.26]	-0.42 [-0.83, -0.09]
	10	-0.59 [-0.99, -0.26]	-0.65 [-1.05, -0.34]	-0.55 [-0.94, -0.25]
	20	-0.99 [-1.30, -0.71]	-0.91 [-1.21, -0.65]	-0.91 [-1.22, -0.64]
	50	-2.00 [-2.30, -1.72]	-1.99 [-2.28, -1.73]	-2.09 [-2.41, -1.80]
	100	-4.40 [-4.80, -4.06]	-4.35 [-4.73, -4.02]	-4.51 [-4.90, -4.17]
Open3DSG	5	-81.70 [-83.66, -79.57]	-82.26 [-84.29, -80.15]	-81.95 [-83.90, -79.84]
	10	-63.78 [-65.82, -61.58]	-63.84 [-65.87, -61.64]	-63.58 [-65.68, -61.33]
	20	-51.40 [-53.18, -49.55]	-51.14 [-52.95, -49.31]	-50.90 [-52.68, -49.01]
	50	-41.48 [-43.11, -39.73]	-41.28 [-42.93, -39.53]	-41.06 [-42.75, -39.28]
	100	-32.39 [-33.82, -30.88]	-32.28 [-33.66, -30.86]	-32.22 [-33.61, -30.74]
SGFN	5	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]
	10	-0.87 [-2.09, 0.23]	-1.08 [-2.24, -0.20]	-0.88 [-1.94, -0.00]
	20	-1.51 [-2.26, -0.89]	-1.46 [-2.19, -0.88]	-1.44 [-2.22, -0.84]
	50	-3.32 [-3.80, -2.90]	-3.35 [-3.83, -2.95]	-3.43 [-3.92, -3.00]
	100	-5.39 [-5.83, -4.96]	-5.37 [-5.79, -4.95]	-5.57 [-6.03, -5.12]

Table 13: Paired point- and mesh-based audit changes for RelCompat3D-Linear. Each entry is Linear minus Source Violation in percentage points with a paired scan-level cluster 95% interval. The SGFN result at $K = 5$ is tied; the point-only SGFN interval at $K = 10$ contains zero.

Predictor	K	Point ΔV (95% CI)	Mesh ΔV (95% CI)	Agreement ΔV (95% CI)
VL-SAT	5	-0.51 [-0.94, -0.17]	-1.01 [-1.85, -0.34]	-0.51 [-0.95, -0.17]
	10	-1.03 [-1.50, -0.64]	-0.99 [-1.47, -0.61]	-0.83 [-1.27, -0.46]
	20	-1.21 [-1.56, -0.89]	-1.19 [-1.54, -0.86]	-1.10 [-1.43, -0.78]
	50	-2.37 [-2.71, -2.06]	-2.31 [-2.65, -2.01]	-2.38 [-2.73, -2.07]
	100	-4.39 [-4.79, -4.05]	-4.33 [-4.71, -4.02]	-4.43 [-4.84, -4.10]
Open3DSG	5	-66.02 [-68.35, -63.30]	-66.37 [-68.53, -63.80]	-66.36 [-68.57, -63.78]
	10	-52.32 [-54.26, -50.28]	-52.11 [-54.11, -50.07]	-52.01 [-54.05, -49.98]
	20	-45.50 [-47.35, -43.60]	-45.00 [-46.74, -43.18]	-44.63 [-46.41, -42.78]
	50	-37.74 [-39.47, -35.99]	-37.58 [-39.38, -35.78]	-37.12 [-38.96, -35.28]
	100	-28.66 [-30.00, -27.26]	-28.67 [-30.07, -27.26]	-28.56 [-29.96, -27.12]
SGFN	5	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]
	10	-1.53 [-2.85, -0.46]	-1.30 [-2.54, -0.39]	-1.11 [-2.27, -0.21]
	20	-2.07 [-2.86, -1.36]	-1.98 [-2.73, -1.36]	-1.78 [-2.54, -1.14]
	50	-3.96 [-4.50, -3.45]	-3.90 [-4.47, -3.39]	-3.93 [-4.48, -3.43]
	100	-5.56 [-6.00, -5.11]	-5.58 [-6.03, -5.13]	-5.67 [-6.15, -5.22]

Table 14: Paired point- and mesh-based audit changes for RelCompat3D-MLP. Each entry is MLP minus Source Violation in percentage points with a paired scan-level cluster 95% interval. The SGFN result at $K = 5$ is tied.

(a) $K = 50$

Condition	Order	VL-SAT	Open3DSG	SGFN
Default	1.000	.9277/.0197	.4418/.0342	.7450/.0263
Proximity 2.0	1.000	.9280/.0197	.4436/.0342	.7470/.0263
Proximity 3.0	1.000	.9275/.0198	.4426/.0342	.7437/.0264
Vertical 0.20 m	1.000	.9277/.0194	.4411/.0342	.7450/.0258
Vertical 0.30 m	1.000	.9277/.0204	.4439/.0342	.7447/.0274
Negative cap 1	1.000	.9277/.0197	.4428/.0342	.7447/.0263
Negative cap 4	1.000	.9277/.0197	.4416/.0342	.7455/.0263
Pair weight .125	1.000	.9277/.0198	.4421/.0342	.7450/.0264
Pair weight .5	1.000	.9277/.0197	.4421/.0342	.7450/.0262

(b) $K = 100$

Condition	Order	VL-SAT	Open3DSG	SGFN
Default	1.000	.9658/.0295	.5685/.0324	.9303/.0350
Proximity 2.0	1.000	.9660/.0294	.5720/.0324	.9325/.0350
Proximity 3.0	1.000	.9653/.0296	.5662/.0324	.9277/.0350
Vertical 0.20 m	1.000	.9658/.0289	.5675/.0324	.9308/.0334
Vertical 0.30 m	1.000	.9658/.0306	.5672/.0324	.9300/.0370
Negative cap 1	1.000	.9658/.0295	.5700/.0324	.9293/.0350
Negative cap 4	1.000	.9658/.0295	.5670/.0324	.9305/.0350
Pair weight .125	1.000	.9658/.0296	.5690/.0324	.9303/.0351
Pair weight .5	1.000	.9658/.0294	.5682/.0324	.9303/.0347

Table 15: Counterfactual-policy sensitivity. Order is linked-positive ordering accuracy over the re-ranked proximity and vertical sensitivity pairs on the 117-scan internal-development split. Metric cells report exact-match Recall / verifier-derived Violation. Only the named factor changes from the default.

Predictor	Estimator	Five smooth mappings			Percentile mapping		
		Fav.	Min. ΔR	Max. ΔV	Fav.	Min. ΔR	Max. ΔV
VL-SAT	Linear	25/25	0.000	-0.036	5/5	+0.025	-0.146
	MLP	24/25	-0.025	-0.109	3/5	-0.201	-0.146
Open3DSG	Linear	25/25	+0.302	-9.163	5/5	+0.277	-8.982
	MLP	25/25	+0.277	-8.240	5/5	+0.201	-7.851
SGFN	Linear	25/25	0.000	0.000	3/5	-0.227	0.000
	MLP	25/25	0.000	0.000	3/5	-0.050	0.000

Table 16: Source-score mapping sensitivity over $K \in \{5, 10, 20, 50, 100\}$. Fav. counts settings with $\Delta R \geq 0$ and $\Delta V \leq 0$ relative to Source. Minimum Recall and maximum Violation changes are percentage points; negative ΔV is preferred. The smooth block pools three power and two logit-temperature mappings.

Predictor	Ranking rule	$K = 5$	$K = 10$	$K = 20$	$K = 50$	$K = 100$
VL-SAT	Source	41.94/0.29	63.22/0.82	80.74/1.42	92.72/2.68	96.35/4.76
	Positive-density	41.14/0.33	62.13/0.97	79.83/1.51	91.77/2.68	96.10/4.46
	Linear	42.07/0.15	63.39/0.57	80.82/1.14	92.77/1.97	96.58/2.95
	MLP	42.09/0.15	63.47/0.51	80.92/1.09	92.72/1.89	96.50/2.96
Open3DSG	Source	3.42/52.05	9.87/32.89	19.89/20.99	40.43/13.87	51.11/12.42
	Positive-density	4.03/2.03	11.56/3.45	22.61/4.30	43.76/5.09	56.92/6.05
	Linear	3.73/0.94	11.38/2.33	23.62/3.13	44.18/3.42	56.85/3.24
	MLP	3.70/4.95	11.78/4.97	24.67/4.56	46.70/4.13	59.89/3.71
SGFN	Source	31.17/2.37	39.75/3.49	49.12/3.22	74.02/3.85	92.35/6.30
	Positive-density	31.14/2.37	39.43/3.58	48.51/3.31	73.87/4.01	91.84/6.19
	Linear	31.17/2.37	39.75/3.47	49.14/2.97	74.50/2.63	93.03/3.50
	MLP	31.17/2.37	39.75/3.47	49.19/2.96	74.57/2.58	92.88/3.50

Table 17: Comparison with the training-positive density baseline on the shared 3DSSG validation target. Cells are exact-match Recall / verifier-derived Violation percentages. Linear and MLP denote the two proposed RelCompat3D estimators under the same family-aware re-ranking rule.

Predictor	K	Hard-tail R/V	Hard-tail U	Hard-drop R/V	Hard-drop U	Selected
VL-SAT	5	41.97/0.15	14.05	41.97/0.00	14.12	2,740
	10	63.34/0.47	14.38	63.17/0.00	14.47	5,480
	20	80.79/0.99	19.00	80.74/0.00	19.20	10,960
	50	92.70/1.61	27.84	92.57/0.00	28.50	27,400
	100	96.37/2.03	35.71	96.27/0.00	37.12	54,781
Open3DSG	5	3.50/0.94	42.06	6.82/0.00	47.47	2,665
	10	10.10/2.33	38.74	13.04/0.00	40.79	5,330
	20	20.39/3.13	36.82	24.12/0.00	39.49	10,660
	50	41.36/3.42	42.31	42.42/0.00	46.36	26,631
	100	52.17/3.23	45.40	53.20/0.00	47.18	52,121
SGFN	5	31.17/2.37	28.98	31.55/0.00	30.00	2,740
	10	39.75/3.47	34.65	40.13/0.00	36.26	5,480
	20	49.12/2.95	37.97	49.65/0.00	39.19	10,960
	50	74.04/2.32	37.15	74.62/0.00	38.07	27,400
	100	92.42/2.29	39.15	92.70/0.00	40.42	54,781

Table 18: Label-consuming direct-verifier diagnostics. R, V, and U are exact-match Recall, primary verifier-derived Violation, and uncertainty percentages. Selected is the total number of Hard-drop outputs across contexts; Hard-tail always fills every available source prefix.

Predictor	Fusion	$K = 50$	$K = 100$
VL-SAT	Linear	.9277/.0197	.9658/.0295
	MLP	.9272/.0189	.9650/.0296
	RankAvg	.9023/.0162	.9705/.0226
	RRF	.9172/.0182	.9705/.0250
Open3DSG	Linear	.4418/.0342	.5685/.0324
	MLP	.4670/.0413	.5989/.0371
	RankAvg	.4320/.0460	.5400/.0549
	RRF	.4245/.0960	.5468/.0780
SGFN	Linear	.7450/.0263	.9303/.0350
	MLP	.7457/.0258	.9288/.0350
	RankAvg	.7022/.0235	.9272/.0260
	RRF	.6850/.0254	.8799/.0307

Table 19: Compatibility-estimator and fusion comparisons. Cells are exact-match Recall / verifier-derived Violation at $K = 50$ and $K = 100$. Every row preserves the source family sequence and support/contact selections.

Condition	Open3DSG Linear	Open3DSG MLP	VL-SAT Linear	VL-SAT MLP	SGFN Linear	SGFN MLP
Full method	**44.18/3.42**	**46.70/4.13**	**92.77/1.97**	**92.72/1.89**	**74.50/2.63**	**74.57/2.58**
Wrong predicate	42.65/21.98	46.78/19.30	90.46/4.97	90.99/5.03	71.58/9.42	71.78/9.67
Wrong pair	38.52/8.50	38.02/9.31	90.56/2.61	91.31/2.62	71.55/3.99	72.10/3.91
Shuffled geometry	38.42/12.93	36.83/12.88	89.17/3.05	90.26/3.00	70.59/4.73	71.15/4.59
Fixed-predicate swap	42.67/21.98	45.52/19.13	90.46/4.97	90.99/4.99	71.58/9.42	71.85/9.65
Distance only	51.16/8.24		81.90/5.34		63.19/10.00	
Compatibility only	42.22/3.42	48.19/3.42	67.65/1.61	77.84/1.61	52.79/2.32	57.10/2.33

Table 20: Matched structural controls for both compatibility estimators at $K = 50$. Cells are exact-match Recall / verifier-derived Violation percentages (↑ / ↓). All conditions use the same candidates and family-aware ranking as the main evaluation. Distance only is common to both estimators.

Condition	Open3DSG Linear	Open3DSG MLP	VL-SAT Linear	VL-SAT MLP	SGFN Linear	SGFN MLP
Full method	**56.85/3.24**	**59.89/3.71**	**96.58/2.95**	**96.50/2.96**	**93.03/3.50**	**92.88/3.50**
Wrong predicate	53.47/20.96	58.89/18.34	94.79/7.95	94.99/8.31	89.93/12.99	90.01/13.52
Wrong pair	49.32/8.36	49.90/8.90	94.96/4.73	95.62/4.71	88.07/6.65	89.05/6.59
Shuffled geometry	48.79/12.42	47.94/12.38	94.11/5.60	94.71/5.49	86.83/8.36	87.84/8.06
Fixed-predicate swap	53.63/20.96	57.43/18.15	94.81/7.95	95.09/8.13	90.03/12.98	90.23/13.35
Distance only	63.22/9.55		89.80/8.09		84.06/12.66	
Compatibility only	56.87/3.24	61.33/3.28	83.89/2.03	90.18/2.05	70.64/2.30	80.99/2.31

Table 21: Structural controls at $K = 100$. Cells are exact-match Recall / verifier-derived Violation percentages (↑ / ↓). All conditions use the same candidates and family-aware ranking as the main evaluation. Distance only is common to both estimators.

Predictor	$K = 50$ Family	$K = 50$ Pooled	$K = 100$ Family	$K = 100$ Pooled
VL-SAT	.9293/.0203	.9300/.0219	.9688/.0325	.9690/.0387
Open3DSG	.4685/.0290	.4718/.0528	.6052/.0340	.6443/.0747
SGFN	.7704/.0258	.7925/.0292	.9413/.0371	.9413/.0464

Table 22: Family-specific and pooled products. Cells are exact-match Recall / verifier-derived Violation at $K = 50$ and $K = 100$; both products apply compatibility to all relation families.

Predictor	Estimator	Route	$K = 5$	$K = 10$	$K = 20$	$K = 50$	$K = 100$
VL-SAT	Linear	Family slots	42.07/0.15	63.39/0.57	80.82/1.14	92.77/1.97	96.58/2.95
	Linear	P/V global	42.04/0.15	63.42/0.55	80.84/1.11	92.72/1.95	96.60/2.94
	MLP	Family slots	42.09/0.15	63.47/0.51	80.92/1.09	92.72/1.89	96.50/2.96
	MLP	P/V global	42.12/0.15	63.42/0.49	80.97/1.06	92.77/1.86	96.53/2.93
Open3DSG	Linear	Family slots	3.73/0.94	11.38/2.33	23.62/3.13	44.18/3.42	56.85/3.24
	Linear	P/V global	7.98/0.94	15.84/2.33	25.33/3.13	44.76/3.39	57.83/3.08
	MLP	Family slots	3.70/4.95	11.78/4.97	24.67/4.56	46.70/4.13	59.89/3.71
	MLP	P/V global	7.48/4.17	14.85/4.77	23.92/4.66	43.78/4.16	56.09/3.68
SGFN	Linear	Family slots	31.17/2.37	39.75/3.47	49.14/2.97	74.50/2.63	93.03/3.50
	Linear	P/V global	31.17/2.37	39.75/3.47	49.19/2.97	74.67/2.61	93.28/3.39
	MLP	Family slots	31.17/2.37	39.75/3.47	49.19/2.96	74.57/2.58	92.88/3.50
	MLP	P/V global	31.17/2.37	39.75/3.47	49.27/2.96	75.25/2.57	93.13/3.37

Table 23: Matched routing-constraint comparison on the shared 3DSSG validation target. Cells are exact-match Recall / verifier-derived Violation percentages. P/V global changes only competition between proximity and vertical-order candidates; support/contact positions and identities remain fixed.

Predictor	Estimator	Route	R@50/V@50	R@100/V@100
VL-SAT	Linear	Family slots	92.77/1.97	96.58/2.95
	Linear	P/V global	92.72/1.95	96.60/2.94
	Linear	Support-order only	92.90/1.98	96.58/2.86
	Linear	All families	92.93/2.03	96.88/3.25
	MLP	Family slots	92.72/1.89	96.50/2.96
	MLP	P/V global	92.77/1.86	96.53/2.93
	MLP	Support-order only	92.98/1.89	96.53/2.87
	MLP	All families	92.75/1.92	96.88/3.17
Open3DSG	Linear	Family slots	44.18/3.42	56.85/3.24
	Linear	P/V global	44.76/3.39	57.83/3.08
	Linear	Support-order only	43.71/4.01	54.46/3.20
	Linear	All families	46.42/2.69	60.05/3.31
	MLP	Family slots	46.70/4.13	59.89/3.71
	MLP	P/V global	43.78/4.16	56.09/3.68
	MLP	Support-order only	41.52/4.60	52.95/3.74
	MLP	All families	48.62/3.60	63.65/4.26
SGFN	Linear	Family slots	74.50/2.63	93.03/3.50
	Linear	P/V global	74.67/2.61	93.28/3.39
	Linear	Support-order only	74.45/2.61	92.88/3.22
	Linear	All families	77.04/2.58	94.13/3.71
	MLP	Family slots	74.57/2.58	92.88/3.50
	MLP	P/V global	75.25/2.57	93.13/3.37
	MLP	Support-order only	74.70/2.60	92.75/3.26
	MLP	All families	77.62/2.49	94.21/3.63

Table 24: Additional routing and scope comparisons at $K = 50, 100$ on the shared 3DSSG validation target. Cells are exact-match Recall / verifier-derived Violation percentages. Family slots and P/V global are the direct routing comparison; support-order only and all families relax additional constraints.

Evaluation	Contexts / GT	Source	RelCompat3D-Linear
Public eligible	533 / 3,899	.5206/.1242	.5791/.0324
Public, full target (main)	548 / 3,972	.5111/.1242	.5685/.0324
Recovered full coverage	548 / 3,972	.5161/.1242	.5735/.0332

Table 25: Open3DSG coverage sensitivity at $K = 100$. Cells are exact-match Recall / verifier-derived Violation. The main evaluation uses the full 3,972-relation denominator and no predictions for the 15 public-pipeline drops.

Predictor	K	Pool cov.	Source	Linear	MLP	Active route	Family slots	Unconstrained
VL-SAT	5	99.72	41.94	42.07	42.09	46.22	52.04	56.72
	10	99.72	63.22	63.39	63.47	70.82	78.15	83.76
	20	99.72	80.74	80.82	80.92	87.66	93.73	97.16
	50	99.72	92.72	92.77	92.72	96.73	99.52	99.72
	100	99.72	96.35	96.58	96.50	98.79	99.72	99.72
Open3DSG	5	79.68	3.42	3.73	3.70	8.21	13.72	50.05
	10	79.68	9.87	11.38	11.78	20.49	35.02	69.89
	20	79.68	19.89	23.62	24.67	40.06	61.33	78.65
	50	79.68	40.43	44.18	46.70	63.72	73.46	79.68
	100	79.68	51.11	56.85	59.89	71.88	77.19	79.68
SGFN	5	99.72	31.17	31.17	31.17	31.24	43.00	56.72
	10	99.72	39.75	39.75	39.75	40.43	48.04	83.76
	20	99.72	49.12	49.14	49.19	53.73	58.11	97.16
	50	99.72	74.02	74.50	74.57	86.05	88.07	99.72
	100	99.72	92.35	93.03	92.88	98.49	99.65	99.72

Table 26: Candidate-pool coverage and exact-match Recall upper bounds on the shared 3DSSG validation target. All values are percentages. Pool cov. is the fraction of the 3,972 ground-truth relation identities present anywhere in the fixed candidate pool. Source, Linear, and MLP are observed rankings; the final three columns are diagnostic oracles with progressively relaxed constraints.

Predictor	K	ΔRecall	95% CI	ΔViolation	95% CI
VL-SAT	5	+0.13	[0.00, 0.26]	-0.15	$[-0.30, -0.03]$
	10	+0.18	[0.05, 0.35]	-0.26	$[-0.45, -0.09]$
	20	+0.08	$[-0.03, 0.19]$	-0.28	$[-0.43, -0.16]$
	50	+0.05	$[-0.05, 0.18]$	-0.70	$[-0.88, -0.55]$
	100	+0.23	[0.03, 0.46]	-1.82	$[-2.00, -1.65]$
Open3DSG	5	+0.30	[0.11, 0.49]	-51.11	$[-54.39, -47.86]$
	10	+1.51	[1.04, 2.00]	-30.56	$[-32.55, -28.51]$
	20	+3.73	[2.96, 4.46]	-17.86	$[-18.97, -16.74]$
	50	+3.75	[2.96, 4.50]	-10.45	$[-11.06, -9.86]$
	100	+5.74	[4.63, 6.86]	-9.18	$[-9.62, -8.69]$
SGFN	5	0.00	[0.00, 0.00]	0.00	[0.00, 0.00]
	10	0.00	[0.00, 0.00]	-0.02	$[-0.06, 0.00]$
	20	+0.03	$[-0.07, 0.14]$	-0.25	$[-0.43, -0.11]$
	50	+0.48	[0.29, 0.69]	-1.22	$[-1.44, -1.01]$
	100	+0.68	[0.42, 0.98]	-2.80	$[-3.05, -2.54]$

Table 27: Paired scan-level cluster intervals for RelCompat3D-Linear. Entries are Linear minus Source in percentage points with 95% intervals.

Predictor	K	ΔRecall	95% CI	ΔViolation	95% CI
VL-SAT	5	+0.15	[0.00, 0.33]	-0.15	$[-0.30, -0.03]$
	10	+0.25	[0.07, 0.46]	-0.31	$[-0.53, -0.11]$
	20	+0.18	[0.05, 0.34]	-0.33	$[-0.49, -0.20]$
	50	0.00	[-0.10, 0.10]	-0.79	$[-0.98, -0.62]$
	100	+0.15	[-0.08, 0.39]	-1.80	$[-1.99, -1.63]$
Open3DSG	5	+0.28	[0.08, 0.48]	-47.09	$[-49.94, -44.04]$
	10	+1.91	[1.31, 2.48]	-27.92	$[-29.77, -25.95]$
	20	+4.78	[3.85, 5.72]	-16.44	$[-17.51, -15.34]$
	50	+6.27	[5.02, 7.49]	-9.74	$[-10.31, -9.13]$
	100	+8.79	[7.28, 10.23]	-8.70	$[-9.13, -8.24]$
SGFN	5	0.00	[0.00, 0.00]	0.00	[0.00, 0.00]
	10	0.00	[0.00, 0.00]	-0.02	[-0.06, 0.00]
	20	+0.08	[-0.05, 0.23]	-0.26	$[-0.43, -0.13]$
	50	+0.55	[0.32, 0.81]	-1.27	$[-1.51, -1.06]$
	100	+0.53	[0.28, 0.79]	-2.79	$[-3.06, -2.54]$

Table 28: Paired scan-level cluster intervals for RelCompat3D-MLP. Entries are MLP minus Source in percentage points with 95% intervals.

K	Source R/V	RelCompat3D-Linear R/V	ΔR/ΔV
5	.0465/.0545	.0698/.0182	+.0233/−.0364
10	.0756/.0818	.1453/.0273	+.0698/−.0545
20	.1453/.1273	.2209/.0364	+.0756/−.0909
50	.2616/.1328	.3140/.0904	+.0523/−.0424
100	.3547/.1967	.3547/.1958	+.0000/−.0010

Table 29: ReplicaSSG/FROSS transfer across all reported K values. R is exact-match Recall over 172 relations. V is verifier-derived Violation. The Linear estimator and family-aware ranking procedure match the main experiment.

Predictor	Family	ΔR	ΔV
VL-SAT	Support/contact	.0000	.0000
VL-SAT	Proximity	+.0040	−.0076
VL-SAT	Vertical order	+.0051	−.0609
Open3DSG	Support/contact	.0000	.0000
Open3DSG	Proximity	+.0968	−.0443
Open3DSG	Vertical order	+.1538	−.3014
SGFN	Support/contact	.0000	.0000
SGFN	Proximity	+.0142	−.0150
SGFN	Vertical order	+.0051	−.0643

Table 30: Family-specific metrics within the selected top-100 predictions. Positive ΔV is a regression. V is verifier-derived Violation.

Predictor	Condition	V_{all}	V_{dec}	U	$V_{u \to v}$
VL-SAT	Source	.0476	.0729	.3464	.3941
VL-SAT	Linear	.0295	.0452	.3473	.3768
Open3DSG	Source	.1242	.2126	.4161	.5402
Open3DSG	Linear	.0324	.0548	.4085	.4409
SGFN	Source	.0630	.1005	.3732	.4362
SGFN	Linear	.0350	.0565	.3803	.4153

Table 31: Uncertainty sensitivity at $K = 100$. Lower is better for all four reported quantities.